

Python Tips and Tricks

The following topics as tips and tricks in Python coding are covered in this notebook:

1. How to make Python coding more readable
2. Use `Context Manager` to manage resources
3. Use `try-except-else-finally` for safe exception handling
4. Use `enumerate` for counting
5. Use `zip` for parallel iteration
6. Use unpacking to assign an object to variables
7. Use `Setattr / Getattr` to dynamically set/get attributes of an object
8. Use `getpass` to solicit password from user
9. Use `python -m` to run a module as a script
10. Use `help / dir` for Docstings and Object Attributes
11. Use `Iterable` for iteration

Python Environment Staging

```
In [1]: import sys
print(f'Python sys.version:\n{sys.version}\n')
print(f'Python sys.executable:\n{sys.executable}\n')
print(f'Python sys.path:\n{sys.path}\n')
```

```
Python sys.version:
3.13.2 | packaged by Anaconda, Inc. | (main, Feb 6 2025, 12:55:35)
[Clang 14.0.6 ]
```

```
Python sys.executable:
/Users/michaelkao/miniconda3/envs/pex/bin/python
```

```
Python sys.path:
['/Users/michaelkao/miniconda3/envs/pex/lib/python313.zip', '/Users/
michaelkao/miniconda3/envs/pex/lib/python3.13', '/Users/michaelkao/m
iniconda3/envs/pex/lib/python3.13/lib-dynload', '', '/Users/michaelk
ao/miniconda3/envs/pex/lib/python3.13/site-packages']
```

```
In [2]: import os
```

```
print(f'Current Working Dir (CWD) path:\n{os.getcwd()}\n')
```

```
Current Working Dir (CWD) path:  
/Users/michaelkao/My_Projects/Python
```

1. How to make Python code more readable

1.1 Use descriptive variable and function names

Choose variable and function names that describe what they store or do. For example, instead of `a = 10`, write `user_age = 10`. Clear names make your code self-explanatory and reduce the need for comments.

```
In [3]: # Bad  
a = 5  
b = 2  
  
# Good  
total = 5  
sub_total = 2  
print("Total: ", total)  
print("Subtotal:", sub_total)
```

```
Total:    5  
Subtotal: 2
```

```
In [4]: # Bad  
def a():  
    return 5  
  
def b():  
    return 2  
  
# Good  
def calculate_total():  
    return 5  
  
def calculate_sub_total():  
    return 2  
  
print("Calling calculate_total()    =", calculate_total())  
print("Calling calculate_sub_total() =", calculate_sub_total())
```

```
Calling calculate_total()    = 5  
Calling calculate_sub_total() = 2
```

1.2 Use ternary conditionals (三元條件運算)

Python 使用三元運算子(ternary conditional operator) 好處是程式碼可以看起來比較簡短，在某些情況 python 使用三元運算子簡化後 code 會變得清爽簡短許多。

C 的寫法:

```
x = condition ? 1 : 0
```

Python 的寫法:

```
x = 1 if condition else 0
```

Python 二元運算的寫法較多行但邏輯相同，如下：

```
if condition:
    x = 1
else:
    x = 0
```

```
In [5]: condition = False

x = 1 if condition else 0

# f-strings for Formatting
# Use f-strings for cleaner, more readable string interpolation.
# They're faster and more elegant than concatenation.

print(f"condition = {condition}")
print(f"x = {x}")

print(f'condition is True') if condition else print('condition is False')
```

```
condition = False
x = 0
condition is False
```

1.3 Use _ (Underscore) for ignored variables or as placeholders

When you have a numeric variable with values that are not easy to read, you can improve readability by using _ to signal it's value in thousands.

```
# Bad
num = 1000000
```

```
# Good
num = 1_000_000
```

Or when you don't need a loop variable, use _ to signal it's intentionally ignored. It's a simple but powerful convention.

```
for _ in range(3):
    print("Hello, World!")
```

```
In [6]: num1 = 10000000000
num2 = 1000000000

total = num1 + num2
print(total)
```

```
10100000000
```

```
In [7]: # improving readability via "_" (no-op)
num1 = 10_000_000_000
num2 = 100_000_000

total = num1 + num2
print(total)

# loop counter not used
for _ in range(3):
    print("Hello World!")
```

```
10100000000
Hello World!
Hello World!
Hello World!
```

```
In [8]: # improving reporting via ',' formatter
print(f'{total:,}')
```

```
10,100,000,000
```

1.4 List comprehensions for creating lists

List comprehensions are a compact way to create lists. They make your code shorter and easier to understand compared to traditional loops.

```
squares = [x * x for x in range(10)]
```

```
In [9]: # populate list with squares of numbers from 0 to 4
squares = [x * x for x in range(10)]
print(f"squares: {squares}")
```

```
squares: [0, 1, 4, 9, 16, 25, 36, 49, 64, 81]
```

```
In [10]: # find even numbers from a list
nums = [1, 2, 3, 4, 5, 6]
evens = [num for num in nums if num % 2 == 0]
print(f"evens: {evens}")
```

```
evens: [2, 4, 6]
```

1.5 Swap two variables without using a temporary variable

C 的寫法:

```
int a = 1;
int b = 2;
```

```
temp = a;
a = b;
b = temp;
```

Python 的寫法:

```
a = 1
b = 2

a, b = b, a
```

```
In [11]: # Swapping variables
a = 1
b = 2
a, b = b, a
print(f"a = {a}, b = {b}")

a = 2, b = 1
```

1.6 String Processing

- Reverse a string
- Check if two strings are anagrams
- Count frequency of characters in a string
- Remove duplicates from a string

```
In [12]: # reverse a string
god = "god"
print(f"reversed god: {god[::-1]}")

stressed = "stressed"
print(f"reversed stressed: {stressed[::-1]}")

reversed god: dog
reversed stressed: desserts
```

```
In [13]: # check if two strings are anagrams
str_1 = "listen"
str_2 = "silent"

print(sorted(str_1) == sorted(str_2))

True
```

```
In [14]: # count frequency of characters in a string
from collections import Counter
print (Counter([1,2,2,2,3,3,3,3,3]))

Counter({3: 5, 2: 3, 1: 1})
```

```
In [15]: # Remove duplicates from a string
def remove_duplicates(input_str):
    # Each character will be a key in the dictionary, which eliminates
    # The values are not important, so we use None
    return ''.join(dict.fromkeys(input_str))

# Example usage
text = "hello world"
result = remove_duplicates(text) # "helo wrd"
print(f"Original: {text}")
print(f"Result: {result}")
```

Original: hello world
Result: helo wrd

1.7 List processing

- Reverse a list
- Remove duplicates from a list
- Sort a list
- Flatten a list of lists

```
In [16]: # reverse the list
my_list = [1, 2, 3, 4, 5]
reversed_list = my_list[::-1]
print(F"reversed_list: {reversed_list}")

# remove duplicates from the list
my_list = [1, 2, 2, 3, 3, 3, 4, 5]
unique_list = list(set(my_list))
print(F"unique_list: {unique_list}")

# sort the list
my_list = [3, 1, 4, 1, 5, 9, 2, 6, 5, 3, 5]
sorted_list = sorted(my_list)
print(F"sorted_list: {sorted_list}")

reversed_list: [5, 4, 3, 2, 1]
unique_list: [1, 2, 3, 4, 5]
sorted_list: [1, 1, 2, 3, 3, 4, 5, 5, 5, 6, 9]
```

```
In [17]: # flatten a list of lists
lists = [[1, 2, 3], [4,5], [6, 7, 8]]
flat_list = [x for row in lists for x in row]
print(f"flat_list: {flat_list}")

flat_list: [1, 2, 3, 4, 5, 6, 7, 8]
```

1.8 Dictionary Processing

- loop through dictionary
- extract sub-dictionary

```
In [18]: student = {
    'name': 'John',
    'age': 25,
    'courses': ['Math', 'ComSci']
}

# loop via items()
print("Loop via items() method:")
for key, value in student.items():
    print(f"\t{key}: {value}")

# 使用 keys() 方法遍歷
print("\nLoop via keys() method:")
for key in student.keys():
```

```
print(f"\t{key}")

# 使用 values() 方法遍歷
print("\nLoop via values() method:")
for value in student.values():
    print(f"\t{value}")
```

Loop via items() method:

```
name: John
age: 25
courses: ['Math', 'ComSci']
```

Loop via keys() method:

```
name
age
courses
```

Loop via values() method:

```
John
25
['Math', 'ComSci']
```

```
In [19]: # Using dictionary comprehension to create sub-dictionary
original_dict = {
    'name': 'John',
    'age': 30,
    'city': 'New York',
    'occupation': 'Engineer',
    'email': 'john@example.com'
}
print(f"original_dict: {original_dict}")

# Keys we want in the sub-dictionary
keys = ['name', 'age', 'city']
sub_dict = {key: original_dict[key] for key in keys if key in original_dict}
print(f"sub_dict: {sub_dict}")
```

```
original_dict: {'name': 'John', 'age': 30, 'city': 'New York', 'occupation': 'Engineer', 'email': 'john@example.com'}
sub_dict: {'name': 'John', 'age': 30, 'city': 'New York'}
```

2. Use Context Manager to manage resources

When we read from or write to a file, we sometimes forget to close it at the end. It certainly waste system resources, such as file object and buffer, once allocated and not being able to reuse.

Python has a context manager via "with" statement to manage the resources for you once you exit the execution of "with"-block. Always use with open() for file operations. It ensures files are closed automatically, preventing leaks or errors.

Here is the sample file, called "test.txt", containing the following:

- 1) This is a test file
- 2) With multiple lines of data...
- 3) Third line
- 4) Fourth line
- 5) Fifth line
- 6) Sixth line
- 7) Seventh line
- 8) Eighth line
- 9) Ninth line
- 10) Tenth line

```
In [20]: # Open File Object Modes:
# f = open("test.txt", "r") # for read
# f = open("test.txt", "w") # for write
# f = open("test.txt", "a") # for append
# f = open("test.txt", "r+") # for read and write

f = open('sample/test.txt', 'r')
print(f'File object:\t {f.name}')
print(f'----> Open Mode:\t {f.mode}\n')

file_contents = f.read()
f.close()
print(f'----> is closed?\t {f.closed}\n')

print(f'read the whole file into a string variable at a time:\n{file_contents}\n')

words = file_contents.split(' ')
print(f'words in the "test.txt" file:\n{words}\n')

word_count = len(words)
print(f'There are {word_count} words in the "test.txt" file.\n')
```



```
File object:    sample/test.txt
----> Open Mode:  r
```

```
----> is closed?  True
```

read the whole file into a string variable at a time:

- 1) Here is a joke
- 2) With 6 lines 45 words, it says:
- 3) Teacher: How old is your father, Henry?
- 4) Henry: He is 6 years old.
- 5) Teacher: What? How is this possible?
- 6) Henry: He became father only when I was born.

words in the "test.txt" file:

```
['1)', 'Here', 'is', 'a', 'joke\n2)', 'With', '6', 'lines', '45', 'w',
ords,', 'it', 'says:\n3)', 'Teacher:', 'How', 'old', 'is', 'your', 'f',
father,', 'Henry?\n4)', 'Henry:', 'He', 'is', '6', 'years', 'old.\n',
5)', 'Teacher:', 'What?', 'How', 'is', 'this', 'possible?\n6)', 'Hen',
ry:', 'He', 'became', 'father', 'only', 'when', 'I', 'was', 'born.\n',
n']
```

There are 40 words in the "test.txt" file.

2.1 Read a file using Context Manager

The common reading methods are:

1. `f.read()`
2. `f.readlines()`
3. `f.readline()`
4. `f.seek(position)`
5. `f.tell()`

And attributes:

1. `f.mode`
2. `f.closed`

```
In [21]: # Reading Files via 'with' context manager
# not only file, but also other resources, like threads

with open('sample/test.txt', 'r') as f:

    print(f'File object:\t {f.name}')
    print(f'----> Open Mode:\t {f.mode}\n')

    # File contents in one string variable:
    file_contents = f.read()
    print(f'The whole file was read into a string variable at a time')
```

```

# Context Manager closes the file automatically once out of the with
print(f'----> Context Manager exits. Is the file closed?\t {f.closed}')

try:
    # read file when closed
    no_contents = f.read()

except ValueError:
    print('f.read(): I/O operation on closed file.\n')
except:
    print('Something else went wrong.\n')

words = file_contents.split(' ')
print(f'View words in the "test.txt" file:\n{words}\n')

word_count = len(words)
print(f'There are {word_count} words in the "test.txt" file.\n')

```

```

File object:      sample/test.txt
----> Open Mode:  r

```

The whole file was read into a string variable at a time:

- 1) Here is a joke
- 2) With 6 lines 45 words, it says:
- 3) Teacher: How old is your father, Henry?
- 4) Henry: He is 6 years old.
- 5) Teacher: What? How is this possible?
- 6) Henry: He became father only when I was born.

```
----> Context Manager exits. Is the file closed?  True
```

```
f.read(): I/O operation on closed file.
```

View words in the "test.txt" file:

```
['1)', 'Here', 'is', 'a', 'joke\n2)', 'With', '6', 'lines', '45', 'w
ords,', 'it', 'says:\n3)', 'Teacher:', 'How', 'old', 'is', 'your', '
father,', 'Henry?\n4)', 'Henry:', 'He', 'is', '6', 'years', 'old.\n
5)', 'Teacher:', 'What?', 'How', 'is', 'this', 'possible?\n6)', 'Hen
ry:', 'He', 'became', 'father', 'only', 'when', 'I', 'was', 'born.\n
n']
```

There are 40 words in the "test.txt" file.

2.2 More examples about file reading methods

```

In [22]: # More reading methods
with open('sample/test.txt', 'r') as f:

    # read file contents in a list of lines:
    f_contents = f.readlines()
    print(f'read lines into a list at a time:\n{f_contents}\n')

    # rewind to the head of the file

```

```

f.seek(0)

# read one line at a time:
f_contents = f.readline()
print(f'read first line into a list at a time:\n{f_contents}')

f_contents = f.readline()
print(f'read 2nd line into a list next time:\n{f_contents}')

# mute "\n" effect in the buffer
f_contents = f.readline()
print(f'read 3rd line into a list, ignore \n:\n{f_contents}', end='')
f_contents = f.readline()
print(f'read 4th line into a list, ignore \n:\n{f_contents}', end='')

# Iterating in a for-loop, line by line each loop till the end of file
for line in f:
    print(f'Iterating in a for-loop, line by line each loop till the end of file')

# rewind to the begining of file (BOF)
f.seek(0)

# Iterating in a while-loop in bytes till the end of file (EOF)
size_to_read = 100
f_contents = f.read(size_to_read)
while len(f_contents) > 0:
    print(f'\nIterating in a while-loop, {size_to_read} bytes each loop')
    f_contents = f.read(size_to_read)

# rewind to the head of the file
f.seek(0)

# read in small chunks, 10 characters per chunk:
size_to_read = 10
f_contents = f.read(size_to_read)
print(f'read {size_to_read} bytes, ignore \n:\n->{f_contents}<-')
print(f'Current position of the file objet = {f.tell()}\n')

f.seek(0)
f_contents = f.read(size_to_read)
print(f'back to BOF, read {size_to_read} bytes, ignore \n:\n->{f_contents}<-')

print(f'Current position of the file objet = {f.tell()}\n')

```

```

read lines into a list at a time:
['1) Here is a joke\n', '2) With 6 lines 45 words, it says:\n', '3)
Teacher: How old is your father, Henry?\n', '4) Henry: He is 6 years
old.\n', '5) Teacher: What? How is this possible?\n', '6) Henry: He
became father only when I was born.\n']

```

```

read first line into a list at a time:
1) Here is a joke

```

```

read 2nd line into a list next time:
2) With 6 lines 45 words, it says:

```

```

read 3rd line into a list, ignore \n:
3) Teacher: How old is your father, Henry?
read 4th line into a list, ignore \n:
4) Henry: He is 6 years old.
Iterating in a for-loop, line by line each loop till the end...
->5) Teacher: What? How is this possible?
<-
Iterating in a for-loop, line by line each loop till the end...
->6) Henry: He became father only when I was born.
<-

```

```

Iterating in a while-loop, 100 bytes each loop till the end...
->1) Here is a joke
2) With 6 lines 45 words, it says:
3) Teacher: How old is your father, Henry?
4) H<-

```

```

Iterating in a while-loop, 100 bytes each loop till the end...
->enry: He is 6 years old.
5) Teacher: What? How is this possible?
6) Henry: He became father only whe<-

```

```

Iterating in a while-loop, 100 bytes each loop till the end...
->n I was born.
<-
read 10 bytes, ignore \n:
->1) Here is<-
Current position of the file objet = 10

```

```

back to BOF, read 10 bytes, ignore \n:
->1) Here is<-
Current position of the file objet = 10

```

2.3 Write a file using Context Manager

```

In [23]: # Write to a file using Context Manager
with open("sample/test2.txt", "w") as f:
    f.write("Testing123")
    f.seek(0)
    f.write("Test987\n")

with open("sample/test2.txt", "r") as f:

```

```
# Iterating line by line till the end of file:
for line in f:
    print(f'Iterating line by line till the end...\n->{line}<-\\n')
```

```
Iterating line by line till the end...
->Test987
<-
Iterating line by line till the end...
->23<-
```

2.4 Copy a file using Context Manager

```
In [24]: # Copy a File to Another:
with open("sample/test.txt", "r") as rf:
    with open("test_copy.txt", "w") as wf:
        for line in rf:
            wf.write(line)

file_read = "test_copy.txt"
with open(file_read, "r") as rf:
    print(f'opening {file_read} file:\n')
    for line in rf:
        print(f'Iterating line by line till the end...\n->{line}<-\\n')

# Copy an image: read/write in binary
with open("sample/Olisan.JPG", "rb") as rf:
    with open("sample/Olisan_copy.JPG", "wb") as wf:
        for line in rf:
            wf.write(line)

# Copy an image: in chunks
with open("sample/Olisan.JPG", "rb") as rf:
    with open("sample/Olisan_copy2.JPG", "wb") as wf:
        chunk_size = 4096
        rf_chunk = rf.read(chunk_size)

        while len(rf_chunk) > 0:
            wf.write(rf_chunk)
            rf_chunk = rf.read(chunk_size)

# Show all the images
from PIL import Image
img = Image.open('sample/Olisan.JPG')
img.show()
img_copy = Image.open('sample/Olisan_copy.JPG')
img_copy.show()
img_copy2 = Image.open('sample/Olisan_copy2.JPG')
img_copy2.show()
```

opening test_copy.txt file:

```
Iterating line by line till the end...
->1) Here is a joke
<-
Iterating line by line till the end...
->2) With 6 lines 45 words, it says:
<-
Iterating line by line till the end...
->3) Teacher: How old is your father, Henry?
<-
Iterating line by line till the end...
->4) Henry: He is 6 years old.
<-
Iterating line by line till the end...
->5) Teacher: What? How is this possible?
<-
Iterating line by line till the end...
->6) Henry: He became father only when I was born.
<-
```

3. Use try-except-else-finally for error handling

Use `try/except` to handle runtime errors gracefully. It prevents your code from crashing unexpectedly.

```
In [25]: try:
        # Code that might raise an exception
        result = 10 / 0
    except ZeroDivisionError:
        # Handle the specific exception
        print("Error: Cannot divide by zero")
    else:
        # Code to run if no exception occurs
        print("else: Division successful")
    finally:
        # Code to run regardless of whether an exception occurred or no
        print("finally: This will always execute")
```

```
Error: Cannot divide by zero
finally: This will always execute
```

4. Use enumerate for counting

When looping through lists, use `enumerate()` instead of manual counters. It automatically gives you both index and value, improving clarity.

```
In [26]: # Old way using counter
names = ['Corey', 'Michael', 'Travis', 'Dave']
```

```
index = 0
for name in names:
    print(index, name)
    index += 1
```

```
0 Corey
1 Michael
2 Travis
3 Dave
```

```
In [27]: # cleaner code with enumerator
names = ['Corey', 'Michael', 'Travis', 'Dave']

for index, name in enumerate(names):
    print(index, name)
```

```
0 Corey
1 Michael
2 Travis
3 Dave
```

```
In [28]: # Another cleaner code with enumerator, starting index from 1
names = ['Corey', 'Michael', 'Travis', 'Dave']

# changing starting index from 1
for index, name in enumerate(names, start=1):
    print(index, name)
```

```
1 Corey
2 Michael
3 Travis
4 Dave
```

5. Use `zip` for multiple objects

Combine multiple lists cleanly with `zip()`. It's perfect for looping through related data side by side.

```
In [29]: names = ['Corey', 'Michael', 'Travis', 'Dave']
heroes = ['Spiderman', 'Superman', 'Deadpool', 'Batman']
for index, name in enumerate(names):
    hero = heroes[index]
    print(f'{name} is actually {hero}')
```

```
Corey is actually Spiderman
Michael is actually Superman
Travis is actually Deadpool
Dave is actually Batman
```

```
In [30]: names = ['Corey', 'Michael', 'Travis', 'Dave']
heroes = ['Spiderman', 'Superman', 'Deadpool', 'Batman']

# cleaner code with zip
for name, hero in zip(names, heroes):
    print(f'{name} is actually {hero}')
```

Corey is actually Spiderman
 Michael is actually Superman
 Travis is actually Deadpool
 Dave is actually Batman

```
In [31]: names = ['Corey', 'Michael', 'Travis', 'Dave']
         heroes = ['Spiderman', 'Superman', 'Deadpool', 'Batman']
         universes = ['Marvel', 'DC', 'Tokyo', 'Taiwan']

         # cleaner code with zip more than 2 lists
         for name, hero, universe in zip(names, heroes, universes):
             print(f'{name} is actually {hero} from {universe}')
```

Corey is actually Spiderman from Marvel
 Michael is actually Superman from DC
 Travis is actually Deadpool from Tokyo
 Dave is actually Batman from Taiwan

6. Use unpacking to assign an object to variables

It allows you to assign values to variables in a single line of code.

```
In [32]: # Basic unpacking with tuple
         a, b = (1, 2)
         print(f"a: {a}")
         print(f"b: {b}")
```

a: 1
 b: 2

```
In [33]: # unpacking a list
         x, y = [10, 20]
         print(f"x: {x}, y: {y}")
```

x: 10, y: 20

```
In [34]: # Unpack in function arguments
         def add(a, b, c):
             return a + b + c

         args = (1, 2, 3)
         print(f"add(*args) = {add(*args)}")
```

add(*args) = 6

```
In [35]: # unpack a string which is an iterable, consisting a sequence of characters
         a, b, c = "abc"
         print(f"a={a}, b={b}, c={c}")
```

a=a, b=b, c=c

```
In [36]: # Unpacking a dictionary, using ** to unpack key-value pairs
         def greet(name, age):
```



```

    print(f"Hello, {name}. You are {age} years old.")

    person = {"name": "John", "age": 30}
    greet(**person)

    # same as
    greet(name="John", age=30)

```

Hello, John. You are 30 years old.

Hello, John. You are 30 years old.

```

In [37]: # unpacking keys or values from a dictionary
         d = {'a': 1, 'b': 2, 'c': 3}

         k1, k2, k3 = d.keys()
         v1, v2, v3 = d.values()

         print(f"k1={k1}, k2={k2}, k3={k3}")
         print(f"v1={v1}, v2={v2}, v3={v3}")

```

k1=a, k2=b, k3=c

v1=1, v2=2, v3=3

```

In [38]: # Use * for extended unpacking (Python3+)
         a, b, *c, d = (1, 2, 3, 4, 5, 6)
         print(a)
         print(b)
         print(c)
         print(d)

```

1

2

[3, 4, 5]

6

```

In [39]: a, b, *_ , d = (1, 2, 3, 4, 5, 6)
         print(a)
         print(b)
         print(d)

```

1

2

6

```

In [40]: # nested unpacking
         coords = [(1,2),(3,4),(5,6)]
         x,y = zip(*coords)
         print(f"x: {x}")
         print(f"y: {y}")

```

x: (1, 3, 5)

y: (2, 4, 6)

```

In [41]: # more complex nested list
         data = ['Michael', (25, 'Engineer')]

         # unpacking
         name, (age, profession) = data

```

```
print(f"name: {name}, age: {age}, profession: {profession}")
```

name: Michael, age: 25, profession: Engineer

7. Use `setattr/getattr` to dynamically set/get attributes of an object

Setting and getting attributes of an object on the fly.

```
In [42]: class Person():
         pass

person = Person()

# setting attributes and values on the fly
person.first = "Michael"
person.last = "Kao"

print(f'{person.first} {person.last}')
```

Michael Kao

```
In [43]: class Person():
         pass

person = Person()

# setting an attribute with the value of a variable on the fly
key = "first"
value = "Michael"
person.last = "Kao"

setattr(person, key, value)

print(f'{person.first} {person.last}')
```

Michael Kao

```
In [44]: class Person():
         pass

person = Person()

# setting an attribute by the value of a variable on the fly
key = "first"
value = "Michael"
person.last = "Kao"

setattr(person, key, value)

# getting an attribute value by the value of a variable
first = getattr(person, key)
print(first)
```

Michael

```
In [45]: class Person():
           pass

person = Person()

person_dict = {'first': 'Michael', 'last': 'Kao'}

for key, value in person_dict.items():
    setattr(person, key, value)

print(f'{person.first} {person.last}')

for key in person_dict.keys():
    print(getattr(person, key))
```

Michael Kao
Michael
Kao

8. Use `getpass` to solicit password from user

Solicit a password from user, and not showing it on terminal

```
In [46]: username = input('Username: ')
password = input('Password: ')

print('Logging In...')
```

Logging In...

```
In [47]: from getpass import getpass

username = input('Username: ')
password = getpass('Password: ')

print('Logging In...')
```

Logging In...

9. Use `Python dash m` to run a module as a script

Spawn a process at the cmd prompt with args

`python -m module_name`

```
In [48]: # Run a command line from Jupyter notebook cell
!python -m pip list
```

Package	Version
absl-py	2.1.0
annotated-doc	0.0.3

annotated-types	0.7.0
anyio	4.9.0
appdirs	1.4.4
appnope	0.1.4
argon2-cffi	25.1.0
argon2-cffi-bindings	21.2.0
arrow	1.3.0
asttokens	3.0.0
async-lru	2.0.5
attrs	25.3.0
babel	2.17.0
beautifulsoup4	4.13.4
bleach	6.2.0
bs4	0.0.2
certifi	2025.4.26
cffi	1.17.1
charset-normalizer	3.4.0
click	8.1.8
comm	0.2.2
contourpy	1.3.1
cycler	0.12.1
debugpy	1.8.11
decorator	5.2.1
defusedxml	0.7.1
dnspython	2.7.0
email_validator	2.2.0
et_xmlfile	2.0.0
exceptiongroup	1.3.0
executing	2.2.0
fastapi	0.121.0
fastjsonschema	2.21.1
fibheap	0.2.1
fonttools	4.57.0
fqdn	1.5.1
glog	0.3.1
graphviz	0.20.3
greenlet	3.2.3
h11	0.16.0
httpcore	1.0.9
httpx	0.28.1
idna	3.10
importlib_metadata	8.7.0
ipykernel	6.29.5
ipython	9.2.0
ipython_pygments_lexers	1.1.1
isoduration	20.11.0
jedi	0.19.2
Jinja2	3.1.6
json5	0.12.0
jsonpointer	3.0.0
jsonschema	4.24.0
jsonschema-specifications	2025.4.1
jupyter_client	8.6.3
jupyter_core	5.9.1
jupyter-events	0.12.0
jupyter-lsp	2.2.5

jupyter_server	2.16.0
jupyter_server_terminals	0.5.3
jupyterlab	4.4.10
jupyterlab_pygments	0.3.0
jupyterlab_server	2.27.3
kiwisolver	1.4.8
MarkupSafe	3.0.2
matplotlib	3.10.1
matplotlib-inline	0.1.7
mistune	3.1.3
MouseInfo	0.1.3
nbclient	0.10.2
nbconvert	7.16.6
nbformat	5.10.4
nest_asyncio	1.6.0
notebook	7.4.3
notebook_shim	0.2.4
numpy	2.1.3
openpyxl	3.1.5
overrides	7.7.0
packaging	24.2
pandas	2.2.3
pandoc	2.4
pandocfilters	1.5.1
parso	0.8.4
pexpect	4.9.0
pickleshare	0.7.5
pillow	11.1.0
pip	25.3
platformdirs	4.3.8
playwright	1.52.0
plumbum	1.9.0
ply	3.11
prometheus_client	0.22.1
prompt_toolkit	3.0.51
psutil	7.0.0
ptyprocess	0.7.0
pure_eval	0.2.3
PyAutoGUI	0.9.54
pycparser	2.22
pydantic	2.11.3
pydantic_core	2.33.1
pyee	13.0.0
pygame	2.6.1
PyGetWindow	0.0.9
Pygments	2.19.1
PyMsgBox	1.0.9
pyobjc-core	11.0
pyobjc-framework-Cocoa	11.0
pyobjc-framework-Quartz	11.0
pyparsing	3.2.3
pyperclip	1.9.0
pyppeteer	2.0.0
PyQt5	5.15.11
PyQt5-Qt5	5.15.17
PyQt5_sip	12.17.0

PyRect	0.2.0
PyScreeze	1.0.1
python-dateutil	2.9.0.post0
python-dotenv	1.1.0
python-gflags	3.1.2
python-json-logger	3.3.0
pytweening	1.2.0
pytz	2024.2
PyYAML	6.0.2
pymq	26.2.0
qrcode	8.1
referencing	0.36.2
requests	2.32.5
rfc3339-validator	0.1.4
rfc3986-validator	0.1.1
rpds-py	0.25.1
rubicon-objc	0.5.0
schedule	1.2.2
Send2Trash	1.8.3
setuptools	80.9.0
six	1.16.0
sniffio	1.3.1
sortedcollections	2.1.0
sortedcontainers	2.4.0
soupsieve	2.7
stack_data	0.6.3
starlette	0.49.3
terminado	0.18.1
tinycss2	1.4.0
tornado	6.5
tqdm	4.67.1
traitlets	5.14.3
types-python-dateutil	2.9.0.20250516
typing_extensions	4.13.2
typing-inspection	0.4.0
tzdata	2024.2
uri-template	1.3.0
urllib3	2.5.0
uv	0.9.7
uvicorn	0.34.1
wcwidth	0.2.13
webcolors	24.11.1
webencodings	0.5.1
websocket-client	1.8.0
websockets	10.4
wheel	0.44.0
xlrd	2.0.1
zipp	3.21.0

10. Use `help/dir` for Docstings and Object Attributes

```
In [49]: import uvicorn
```

```
# query online document
help(uvicorn)
```

Help on package uvicorn:

NAME

uvicorn

PACKAGE CONTENTS

```
__main__
_subprocess
_types
config
importer
lifespan (package)
logging
loops (package)
main
middleware (package)
protocols (package)
server
supervisors (package)
workers
```

CLASSES

```
builtins.object
    uvicorn.config.Config
    uvicorn.server.Server
```

```
class Config(builtins.object)
|   Config(
|       app: 'ASGIApplication | Callable[..., Any] | str',
|       host: 'str' = '127.0.0.1',
|       port: 'int' = 8000,
|       uds: 'str | None' = None,
|       fd: 'int | None' = None,
|       loop: 'LoopSetupType' = 'auto',
|       http: 'type[asyncio.Protocol] | HTTPProtocolType' = 'auto',
|       ws: 'type[asyncio.Protocol] | WSProtocolType' = 'auto',
|       ws_max_size: 'int' = 16777216,
|       ws_max_queue: 'int' = 32,
|       ws_ping_interval: 'float | None' = 20.0,
|       ws_ping_timeout: 'float | None' = 20.0,
|       ws_per_message_deflate: 'bool' = True,
|       lifespan: 'LifespanType' = 'auto',
|       env_file: 'str | os.PathLike[str] | None' = None,
|       log_config: 'dict[str, Any] | str | RawConfigParser | I
o[Any] | None' = {'version': 1, 'disable_existing_loggers': False, '
formatters': {'default': {'()': 'uvicorn.logging.DefaultFormatter',
'fmt': '%(levelprefix)s %(message)s', 'use_colors': None}, 'access':
{'()': 'uvicorn.logging.AccessFormatter', 'fmt': '%(levelprefix)s %(
client_addr)s - "%(request_line)s" %(status_code)s'}}, 'handlers':
{'default': {'formatter': 'default', 'class': 'logging.StreamHandle
r', 'stream': 'ext://sys.stderr'}, 'access': {'formatter': 'access',
'class': 'logging.StreamHandler', 'stream': 'ext://sys.stdout'}}}, 'l
```

```

oggers': {'uvicorn': {'handlers': ['default'], 'level': 'INFO', 'propagate': False}, 'uvicorn.error': {'level': 'INFO'}, 'uvicorn.access': {'handlers': ['access'], 'level': 'INFO', 'propagate': False}},
    |     log_level: 'str | int | None' = None,
    |     access_log: 'bool' = True,
    |     use_colors: 'bool | None' = None,
    |     interface: 'InterfaceType' = 'auto',
    |     reload: 'bool' = False,
    |     reload_dirs: 'list[str] | str | None' = None,
    |     reload_delay: 'float' = 0.25,
    |     reload_includes: 'list[str] | str | None' = None,
    |     reload_excludes: 'list[str] | str | None' = None,
    |     workers: 'int | None' = None,
    |     proxy_headers: 'bool' = True,
    |     server_header: 'bool' = True,
    |     date_header: 'bool' = True,
    |     forwarded_allow_ips: 'list[str] | str | None' = None,
    |     root_path: 'str' = '',
    |     limit_concurrency: 'int | None' = None,
    |     limit_max_requests: 'int | None' = None,
    |     backlog: 'int' = 2048,
    |     timeout_keep_alive: 'int' = 5,
    |     timeout_notify: 'int' = 30,
    |     timeout_graceful_shutdown: 'int | None' = None,
    |     callback_notify: 'Callable[..., Awaitable[None]] | None'
= None,
    |     ssl_keyfile: 'str | os.PathLike[str] | None' = None,
    |     ssl_certfile: 'str | os.PathLike[str] | None' = None,
    |     ssl_keyfile_password: 'str | None' = None,
    |     ssl_version: 'int' = <SSLMethod.PROTOCOL_TLS_SERVER: 17
>,
    |     ssl_cert_reqs: 'int' = <VerifyMode.CERT_NONE: 0>,
    |     ssl_ca_certs: 'str | None' = None,
    |     ssl_ciphers: 'str' = 'TLSv1',
    |     headers: 'list[tuple[str, str]] | None' = None,
    |     factory: 'bool' = False,
    |     h11_max_incomplete_event_size: 'int | None' = None
    | )
    |
    | Methods defined here:
    |
    | __init__(
    |     self,
    |     app: 'ASGIApplication | Callable[..., Any] | str',
    |     host: 'str' = '127.0.0.1',
    |     port: 'int' = 8000,
    |     uds: 'str | None' = None,
    |     fd: 'int | None' = None,
    |     loop: 'LoopSetupType' = 'auto',
    |     http: 'type[asyncio.Protocol] | HTTPProtocolType' = 'auto',
o',
    |     ws: 'type[asyncio.Protocol] | WSProtocolType' = 'auto',
    |     ws_max_size: 'int' = 16777216,
    |     ws_max_queue: 'int' = 32,
    |     ws_ping_interval: 'float | None' = 20.0,
    |     ws_ping_timeout: 'float | None' = 20.0,

```



```

        ws_per_message_deflate: 'bool' = True,
        lifespan: 'LifespanType' = 'auto',
        env_file: 'str | os.PathLike[str] | None' = None,
        log_config: 'dict[str, Any] | str | RawConfigParser | I
0[Any] | None' = {'version': 1, 'disable_existing_loggers': False, '
formatters': {'default': {'()': 'uvicorn.logging.DefaultFormatter',
'fmt': '%(levelprefix)s %(message)s', 'use_colors': None}, 'access':
{'()': 'uvicorn.logging.AccessFormatter', 'fmt': '%(levelprefix)s %(
client_addr)s - "%(request_line)s" %(status_code)s'}}, 'handlers':
{'default': {'formatter': 'default', 'class': 'logging.StreamHandle
r', 'stream': 'ext://sys.stderr'}, 'access': {'formatter': 'access',
'class': 'logging.StreamHandler', 'stream': 'ext://sys.stdout'}}, 'l
oggers': {'uvicorn': {'handlers': ['default'], 'level': 'INFO', 'pro
propagate': False}, 'uvicorn.error': {'level': 'INFO'}, 'uvicorn.acces
s': {'handlers': ['access'], 'level': 'INFO', 'propagate': False}},
        log_level: 'str | int | None' = None,
        access_log: 'bool' = True,
        use_colors: 'bool | None' = None,
        interface: 'InterfaceType' = 'auto',
        reload: 'bool' = False,
        reload_dirs: 'list[str] | str | None' = None,
        reload_delay: 'float' = 0.25,
        reload_includes: 'list[str] | str | None' = None,
        reload_excludes: 'list[str] | str | None' = None,
        workers: 'int | None' = None,
        proxy_headers: 'bool' = True,
        server_header: 'bool' = True,
        date_header: 'bool' = True,
        forwarded_allow_ips: 'list[str] | str | None' = None,
        root_path: 'str' = '',
        limit_concurrency: 'int | None' = None,
        limit_max_requests: 'int | None' = None,
        backlog: 'int' = 2048,
        timeout_keep_alive: 'int' = 5,
        timeout_notify: 'int' = 30,
        timeout_graceful_shutdown: 'int | None' = None,
        callback_notify: 'Callable[..., Awaitable[None]] | None'
= None,
        ssl_keyfile: 'str | os.PathLike[str] | None' = None,
        ssl_certfile: 'str | os.PathLike[str] | None' = None,
        ssl_keyfile_password: 'str | None' = None,
        ssl_version: 'int' = <SSLMethod.PROTOCOL_TLS_SERVER: 17
>,
        ssl_cert_reqs: 'int' = <VerifyMode.CERT_NONE: 0>,
        ssl_ca_certs: 'str | None' = None,
        ssl_ciphers: 'str' = 'TLSv1',
        headers: 'list[tuple[str, str]] | None' = None,
        factory: 'bool' = False,
        h11_max_incomplete_event_size: 'int | None' = None
    )
    Initialize self. See help(type(self)) for accurate sign
ature.

    bind_socket(self) -> 'socket.socket'

    configure_logging(self) -> 'None'

```

```

| load(self) -> 'None'
|
| setup_event_loop(self) -> 'None'
|
| -----
|
| Readonly properties defined here:
|
| asgi_version
|
| is_ssl
|
| should_reload
|
| use_subprocess
|
| -----
|
| Data descriptors defined here:
|
| __dict__
|     dictionary for instance variables
|
| __weakref__
|     list of weak references to the object
|
class Server(builtins.object)
| Server(config: 'Config') -> 'None'
|
| Methods defined here:
|
| __init__(self, config: 'Config') -> 'None'
|     Initialize self. See help(type(self)) for accurate signature.
|
| capture_signals(self) -> 'Generator[None, None, None]'
|
| handle_exit(self, sig: 'int', frame: 'FrameType | None') ->
'None'
|
| async main_loop(self) -> 'None'
|
| async on_tick(self, counter: 'int') -> 'bool'
|
| run(self, sockets: 'list[socket.socket] | None' = None) -> '
None'
|
| async serve(self, sockets: 'list[socket.socket] | None' = No
ne) -> 'None'
|
| async shutdown(self, sockets: 'list[socket.socket] | None' =
None) -> 'None'
|
| async startup(self, sockets: 'list[socket.socket] | None' =
None) -> 'None'

```

```

|
| -----
|
| Data descriptors defined here:
|
|   __dict__
|       dictionary for instance variables
|
|   __weakref__
|       list of weak references to the object

```

FUNCTIONS

```

run(
    app: 'ASGIApplication | Callable[..., Any] | str',
    *,
    host: 'str' = '127.0.0.1',
    port: 'int' = 8000,
    uds: 'str | None' = None,
    fd: 'int | None' = None,
    loop: 'LoopSetupType' = 'auto',
    http: 'type[asyncio.Protocol] | HTTPProtocolType' = 'auto',
    ws: 'type[asyncio.Protocol] | WSProtocolType' = 'auto',
    ws_max_size: 'int' = 16777216,
    ws_max_queue: 'int' = 32,
    ws_ping_interval: 'float | None' = 20.0,
    ws_ping_timeout: 'float | None' = 20.0,
    ws_per_message_deflate: 'bool' = True,
    lifespan: 'LifespanType' = 'auto',
    interface: 'InterfaceType' = 'auto',
    reload: 'bool' = False,
    reload_dirs: 'list[str] | str | None' = None,
    reload_includes: 'list[str] | str | None' = None,
    reload_excludes: 'list[str] | str | None' = None,
    reload_delay: 'float' = 0.25,
    workers: 'int | None' = None,
    env_file: 'str | os.PathLike[str] | None' = None,
    log_config: 'dict[str, Any] | str | RawConfigParser | IO[Any] | None' = {'version': 1, 'disable_existing_loggers': False, 'formatters': {'default': {'()': 'uvicorn.logging.DefaultFormatter', 'fmt': '%(levelprefix)s %(message)s', 'use_colors': None}, 'access': {'()': 'uvicorn.logging.AccessFormatter', 'fmt': '%(levelprefix)s %(client_addr)s - "%(request_line)s" %(status_code)s'}}, 'handlers': {'default': {'formatter': 'default', 'class': 'logging.StreamHandler', 'stream': 'ext://sys.stderr'}, 'access': {'formatter': 'access', 'class': 'logging.StreamHandler', 'stream': 'ext://sys.stdout'}}, 'loggers': {'uvicorn': {'handlers': ['default'], 'level': 'INFO', 'propagate': False}, 'uvicorn.error': {'level': 'INFO'}, 'uvicorn.access': {'handlers': ['access'], 'level': 'INFO', 'propagate': False}}, 'log_level': 'str | int | None' = None, 'access_log': 'bool' = True, 'proxy_headers': 'bool' = True, 'server_header': 'bool' = True, 'date_header': 'bool' = True, 'forwarded_allow_ips': 'list[str] | str | None' = None, 'root_path': 'str' = '', 'limit_concurrency': 'int | None' = None,

```

```

backlog: 'int' = 2048,
limit_max_requests: 'int | None' = None,
timeout_keep_alive: 'int' = 5,
timeout_graceful_shutdown: 'int | None' = None,
ssl_keyfile: 'str | os.PathLike[str] | None' = None,
ssl_certfile: 'str | os.PathLike[str] | None' = None,
ssl_keyfile_password: 'str | None' = None,
ssl_version: 'int' = <_SSLMethod.PROTOCOL_TLS_SERVER: 17>,
ssl_cert_reqs: 'int' = <VerifyMode.CERT_NONE: 0>,
ssl_ca_certs: 'str | None' = None,
ssl_ciphers: 'str' = 'TLSv1',
headers: 'list[tuple[str, str]] | None' = None,
use_colors: 'bool | None' = None,
app_dir: 'str | None' = None,
factory: 'bool' = False,
h11_max_incomplete_event_size: 'int | None' = None
) -> 'None'

```

DATA

```

__all__ = ['main', 'run', 'Config', 'Server']
main = <Command main>

```

VERSION

```
0.34.1
```

FILE

```

/Users/michaelkao/miniconda3/envs/pex/lib/python3.13/site-packag
es/uvicorn/__init__.py

```

```

In [50]: import uvicorn
         # query object info
         dir(uvicorn)

```

```
Out[50]: ['Config',
          'Server',
          '__all__',
          '__builtins__',
          '__cached__',
          '__doc__',
          '__file__',
          '__loader__',
          '__name__',
          '__package__',
          '__path__',
          '__spec__',
          '__version__',
          '_subprocess',
          '_types',
          'config',
          'importer',
          'logging',
          'main',
          'middleware',
          'run',
          'server',
          'supervisors']
```

```
In [51]: import uvicorn
         # simply use a "?" after an object to find the associated file dire
         uvicorn?
```

```
Type:          module
String form: <module 'uvicorn' from '/Users/michaelkao/miniconda3/en
vs/pex/lib/python3.13/site-packages/uvicorn/__init__.py'>
File:          ~/miniconda3/envs/pex/lib/python3.13/site-packages/uvic
orn/__init__.py
Docstring:     <no docstring>
```

```
In [52]: from datetime import datetime

         # finding attributes and methods of an function object
         dir(datetime)
```

```
Out[52]: ['__add__',
          '__class__',
          '__delattr__',
          '__dir__',
          '__doc__',
          '__eq__',
          '__format__',
          '__ge__',
          '__getattr__',
          '__getstate__',
          '__gt__',
          '__hash__',
          '__init__',
          '__init_subclass__',
          '__le__',
          '__lt__',
```

```

'__ne__',
'__new__',
'__radd__',
'__reduce__',
'__reduce_ex__',
'__replace__',
'__repr__',
'__rsub__',
'__setattr__',
'__sizeof__',
'__str__',
'__sub__',
'__subclasshook__',
'astimezone',
'combine',
'ctime',
'date',
'day',
'dst',
'fold',
'fromisocalendar',
'fromisoformat',
'fromordinal',
'fromtimestamp',
'hour',
'isocalendar',
'isoformat',
'isoweekday',
'max',
'microsecond',
'min',
'minute',
'month',
'now',
'replace',
'resolution',
'second',
'strftime',
'strptime',
'time',
'timestamp',
'timetuple',
'timetz',
'today',
'toordinal',
'tzinfo',
'tzname',
'utcfromtimestamp',
'utcnow',
'utcoffset',
'utctimetuple',
'weekday',
'year']

```

```
In [53]: from datetime import datetime
```

```
# what is it?
print(datetime.today)

# running it
print(datetime.today())
```

```
<built-in method today of type object at 0x105c588b0>
2025-11-10 11:29:59.032450
```

11. Use Iterables for iteration

Iterable is an object, that one can iterate over. It generates an Iterator when passed to `iter()` method.

An iterator is an object, which is used to iterate over an iterable object using the **next()** method.

Iterators have the **next()** method, which returns the next item of the object. Note that every iterator is also an iterable, but not every iterable is an iterator.

For example, a list is iterable but a list is not an iterator. An iterator can be created from an iterable by using the function `iter()`. To make this possible, the class of an object needs either a method **iter**, which returns an iterator, or a **getitem** method with sequential indexes starting with 0.

```
In [54]: for element in ['c123', 142_857, 'Michael', 'Kao']:
          print(element, end=" ")
          print()

          for element in ('c123', 142_857, 'Michael', 'Kao'):
              print(element, end=" ")
              print()

          for letter in 'Michael Kao':
              print(letter, end=" ")
              print()

          for byte in b'Binary':
              print(byte, end=" ")
              print()

          # All objects, shown above, are iterables
          # However, the integer type is not iterable, as follows
          c = 142_857

          # must convert to a string to make integer iterable
          for digit in str(c):
              print(digit, end=" ")
              print()

          # or convert to a list to make integer iterable
          digits = [int(d) for d in str(c)]
```

```

for digit in digits:
    print(digit, end=" ")
print()

# make a deck of cards iterable?
import itertools
ranks = list(range(2, 11)) + ["J", "Q", "K", "A"]
ranks = [str(rank) for rank in ranks]

print(f'Ranks: {ranks}')

suits = ["Spades", "Hearts", "Dimonds", "Clubs"]
deck = [card for card in itertools.product(suits, ranks)]

for (index, card) in enumerate(deck):
    print(f'{1+index}: {card}')

hands = [hand for hand in itertools.combinations(deck, 5)]
print(f'The number of 5-card poker has {len(hands):,} combinations.

```

c123 142857 Michael Kao

c123 142857 Michael Kao

M i c h a e l K a o

66 105 110 97 114 121

1 4 2 8 5 7

1 4 2 8 5 7

Ranks: ['2', '3', '4', '5', '6', '7', '8', '9', '10', 'J', 'Q', 'K', 'A']

1: ('Spades', '2')

2: ('Spades', '3')

3: ('Spades', '4')

4: ('Spades', '5')

5: ('Spades', '6')

6: ('Spades', '7')

7: ('Spades', '8')

8: ('Spades', '9')

9: ('Spades', '10')

10: ('Spades', 'J')

11: ('Spades', 'Q')

12: ('Spades', 'K')

13: ('Spades', 'A')

14: ('Hearts', '2')

15: ('Hearts', '3')

16: ('Hearts', '4')

17: ('Hearts', '5')

18: ('Hearts', '6')

19: ('Hearts', '7')

20: ('Hearts', '8')

21: ('Hearts', '9')

22: ('Hearts', '10')

23: ('Hearts', 'J')

24: ('Hearts', 'Q')

25: ('Hearts', 'K')

26: ('Hearts', 'A')

27: ('Dimonds', '2')

28: ('Dimonds', '3')


```
29: ('Dimonds', '4')
30: ('Dimonds', '5')
31: ('Dimonds', '6')
32: ('Dimonds', '7')
33: ('Dimonds', '8')
34: ('Dimonds', '9')
35: ('Dimonds', '10')
36: ('Dimonds', 'J')
37: ('Dimonds', 'Q')
38: ('Dimonds', 'K')
39: ('Dimonds', 'A')
40: ('Clubs', '2')
41: ('Clubs', '3')
42: ('Clubs', '4')
43: ('Clubs', '5')
44: ('Clubs', '6')
45: ('Clubs', '7')
46: ('Clubs', '8')
47: ('Clubs', '9')
48: ('Clubs', '10')
49: ('Clubs', 'J')
50: ('Clubs', 'Q')
51: ('Clubs', 'K')
52: ('Clubs', 'A')
```

The number of 5-card poker has 2,598,960 combinations.